Assignment 02

Submitted By,

Sreejata Bhattacharjee

Roll no: 25CEM5R10

Submitted To,

Dr Narayanarao

Thermal, Microwave and hyperspectral remote sensing

Derivation of Spectral reflectance curve of IIRS Calibrated Data for CHANDRAYAAN 2 Mission

Aim: Deriving spectral reflectabce curve using lunar data

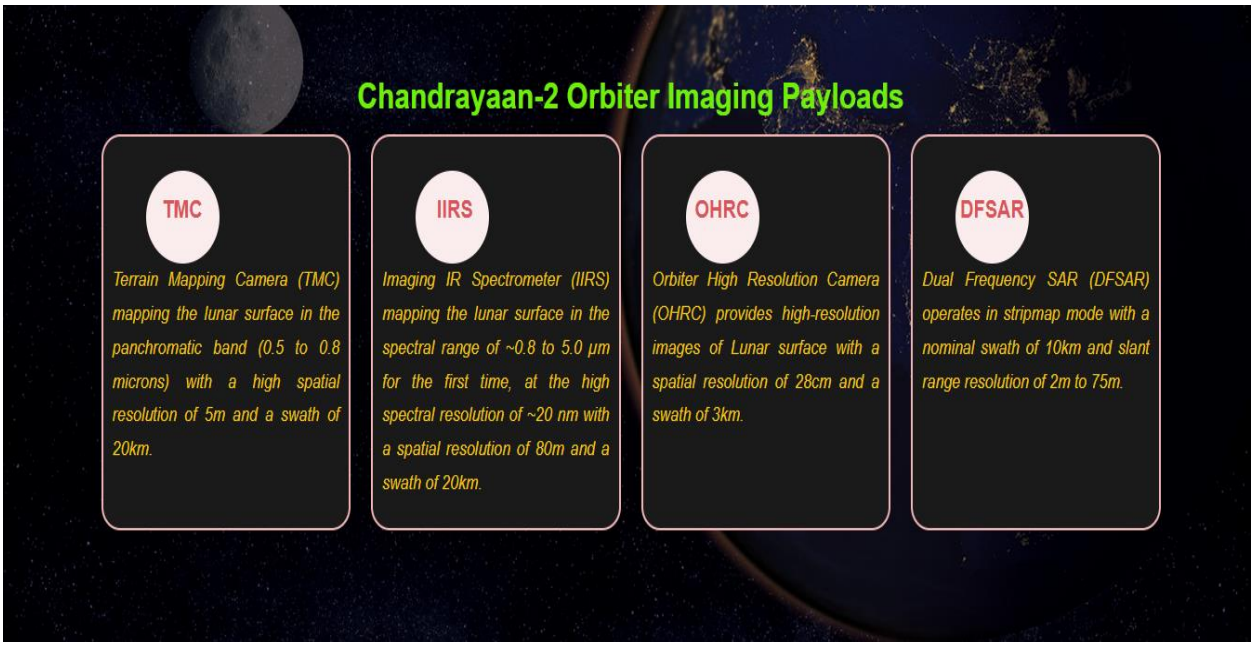
Data Recuirements:I

- IRS Calibrated Data
- Instrument data
- Place of study
- Date of observation
- PDS/Product ID

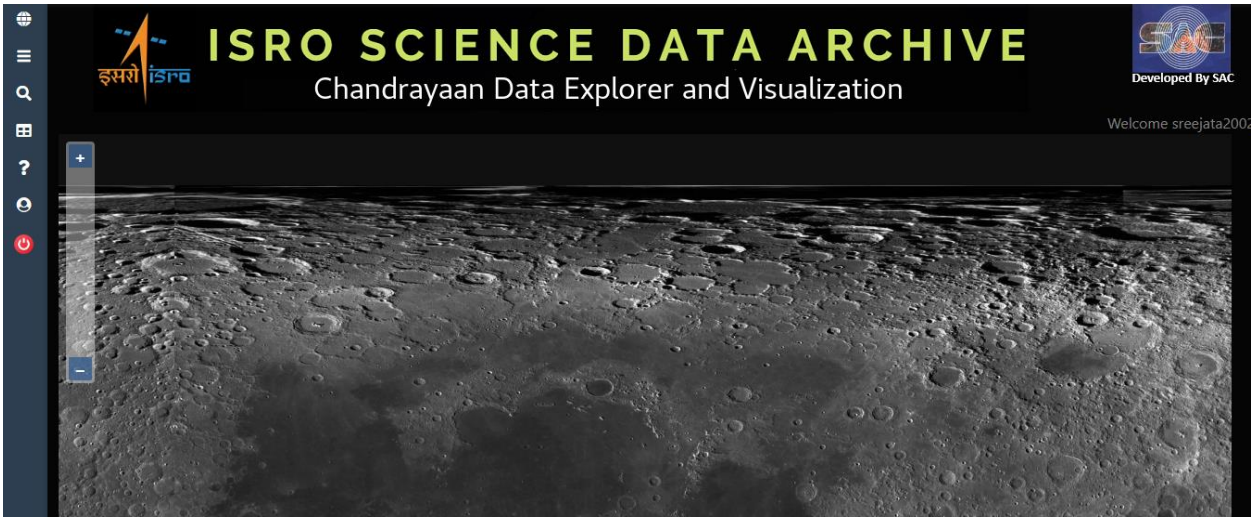
Data sources: https://chmapbrowse.issdc.gov.in/

Software used:

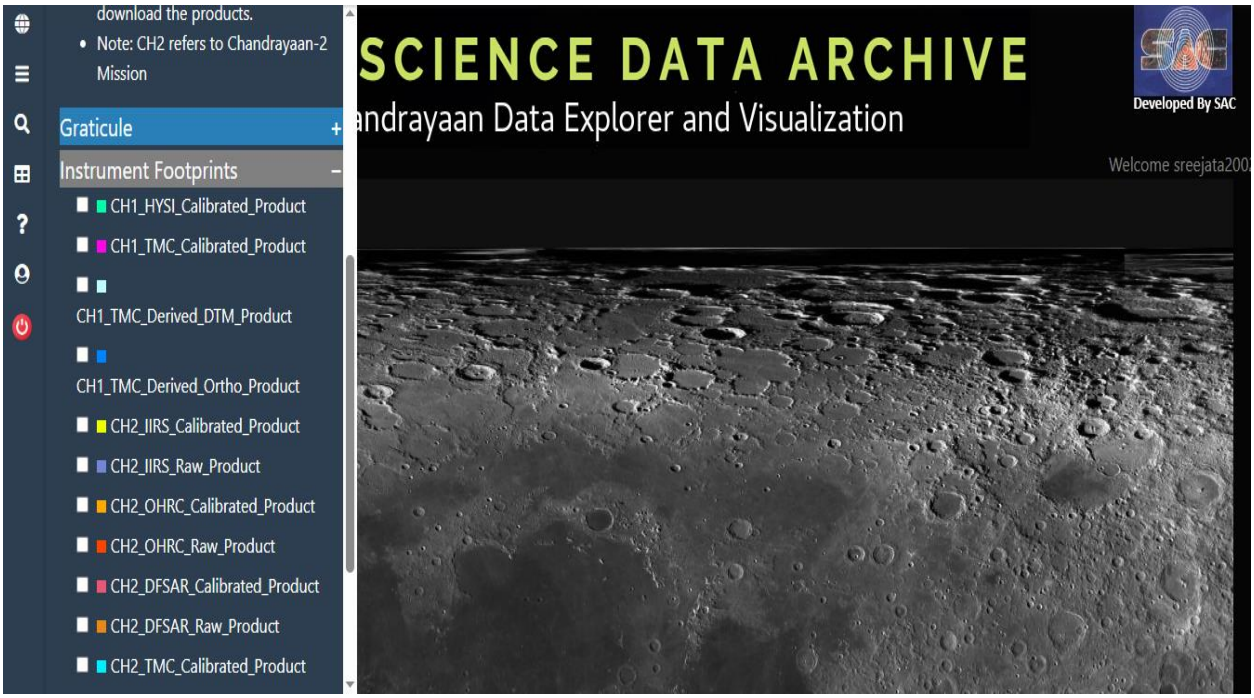
- Python(spectral curve code)
- Jupyter lab(compiler platform)
- Jupyter notebook(compiler)



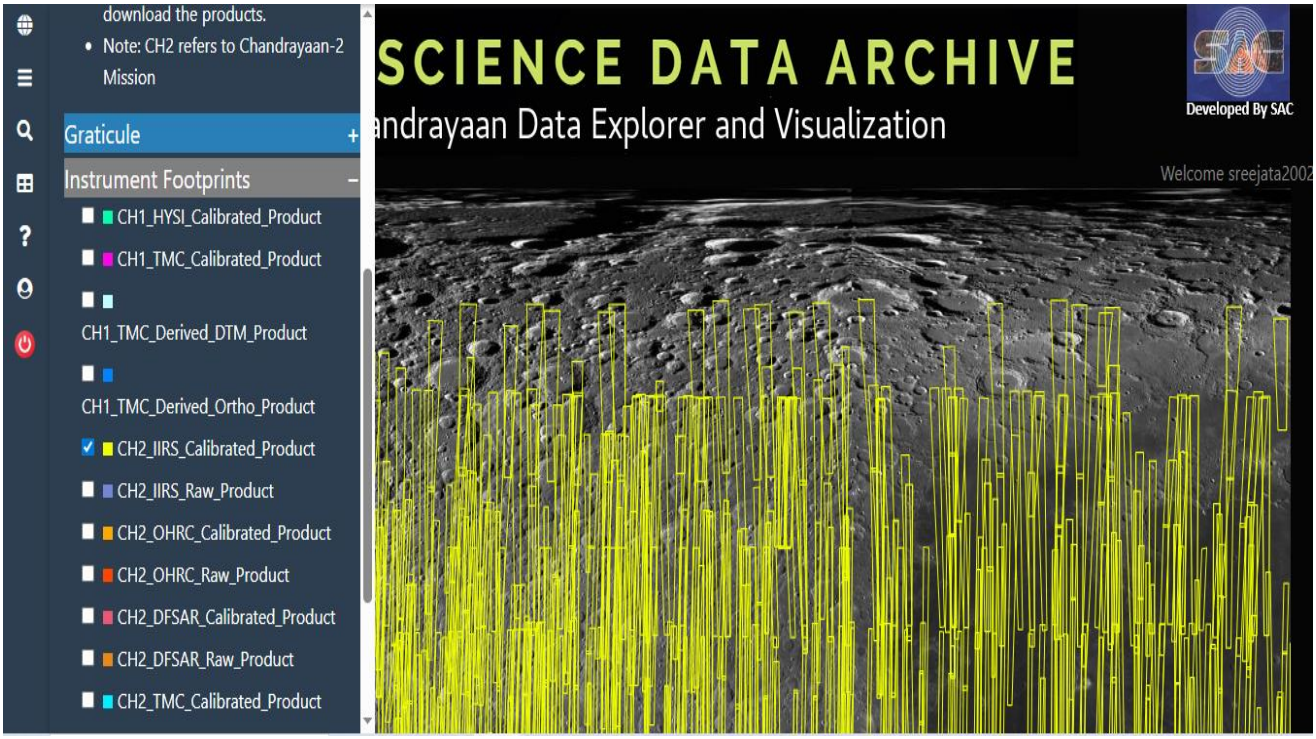
Display of CHANDRAYAAN 2 Imaging payloads



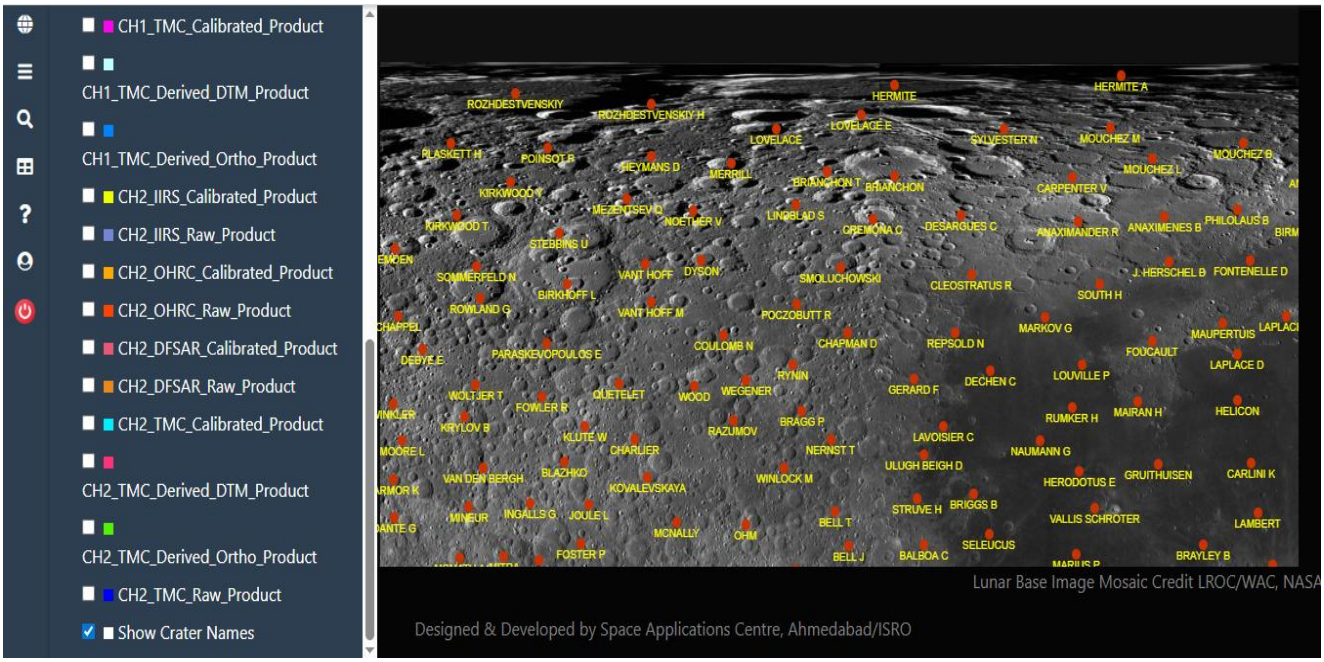
Display of Chandrayaan data explorer



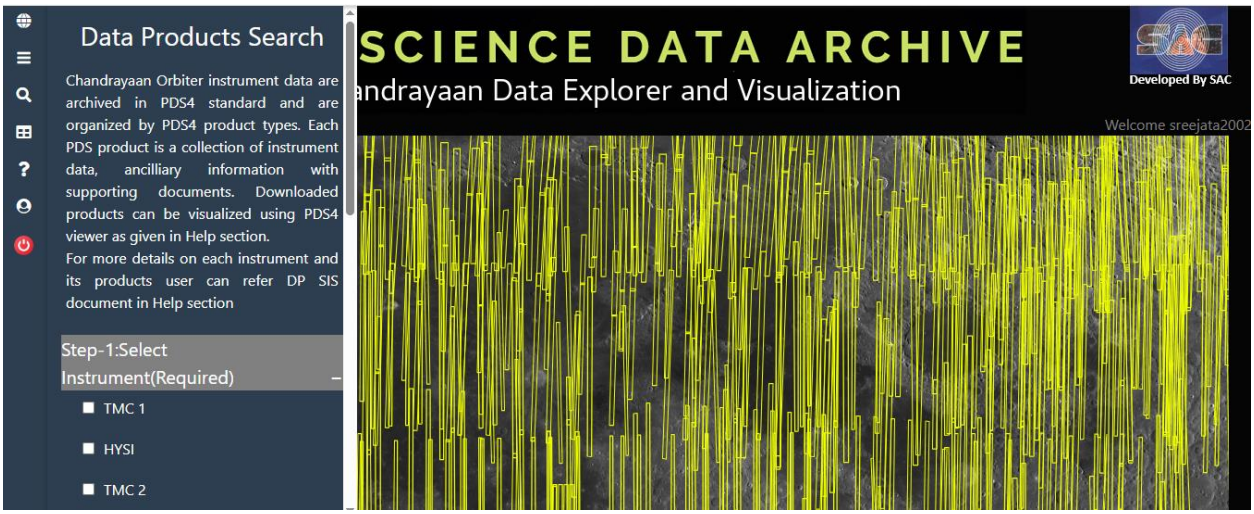
Display of CHANDRAYAAN 2 Instrument footprints



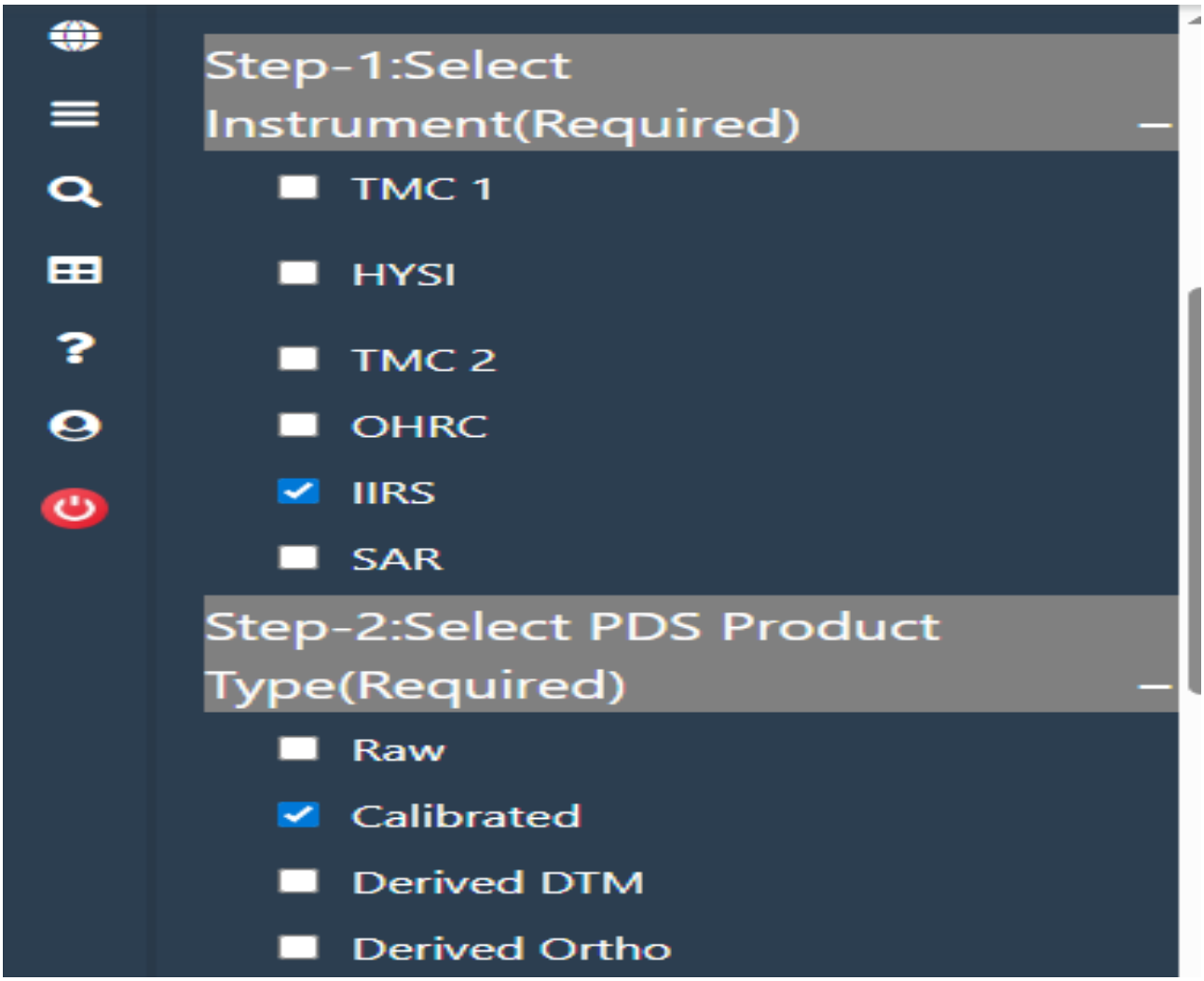
Display of CHANDRAYAAN 2 product review(Ch2 Calibrated product)



Display of names of the craters on lunar surface

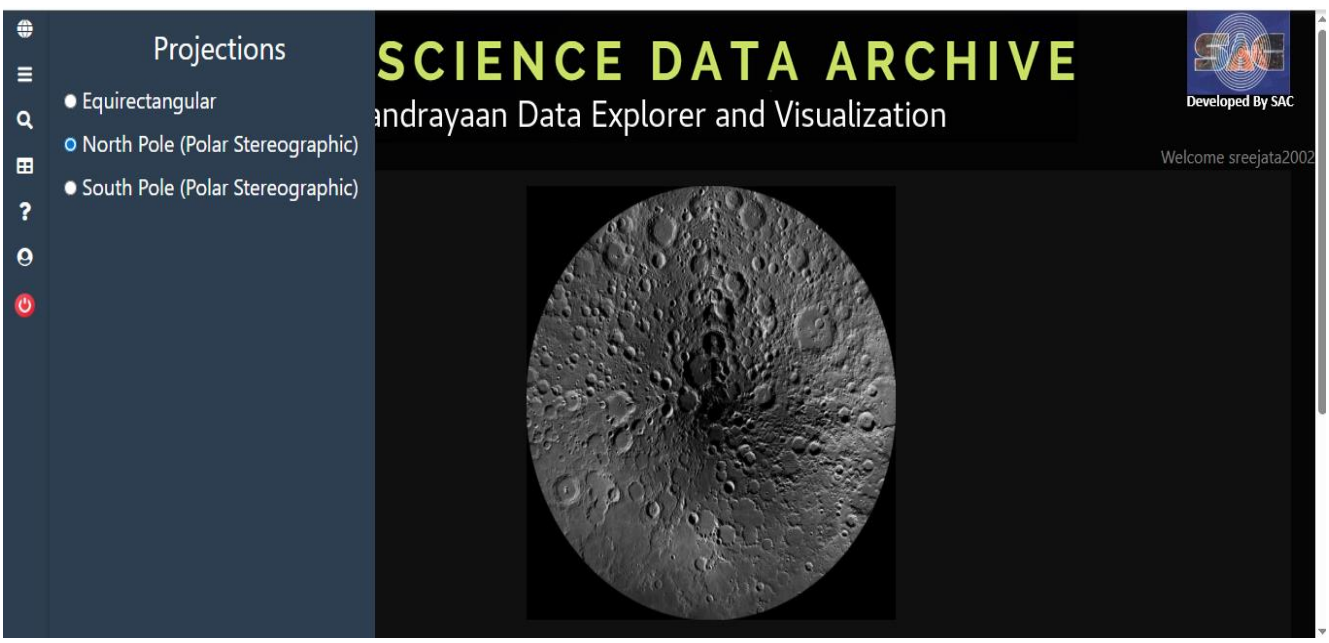


Selection of the Instrument(IIRS)

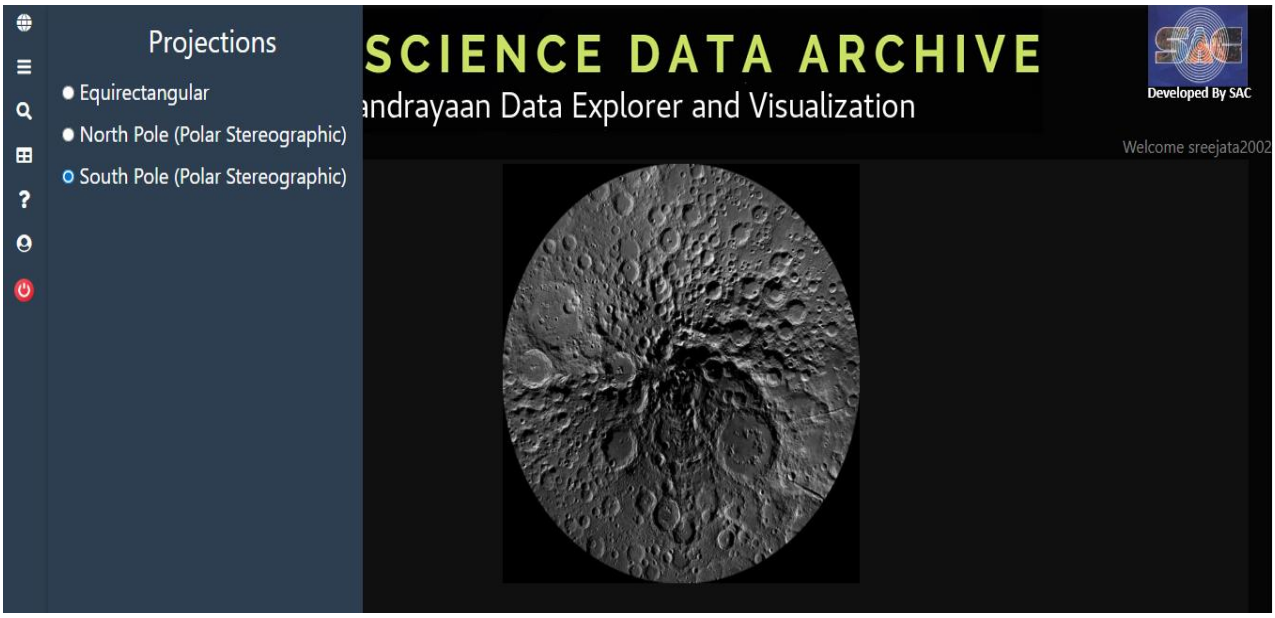


Selection of the PDS Product typ

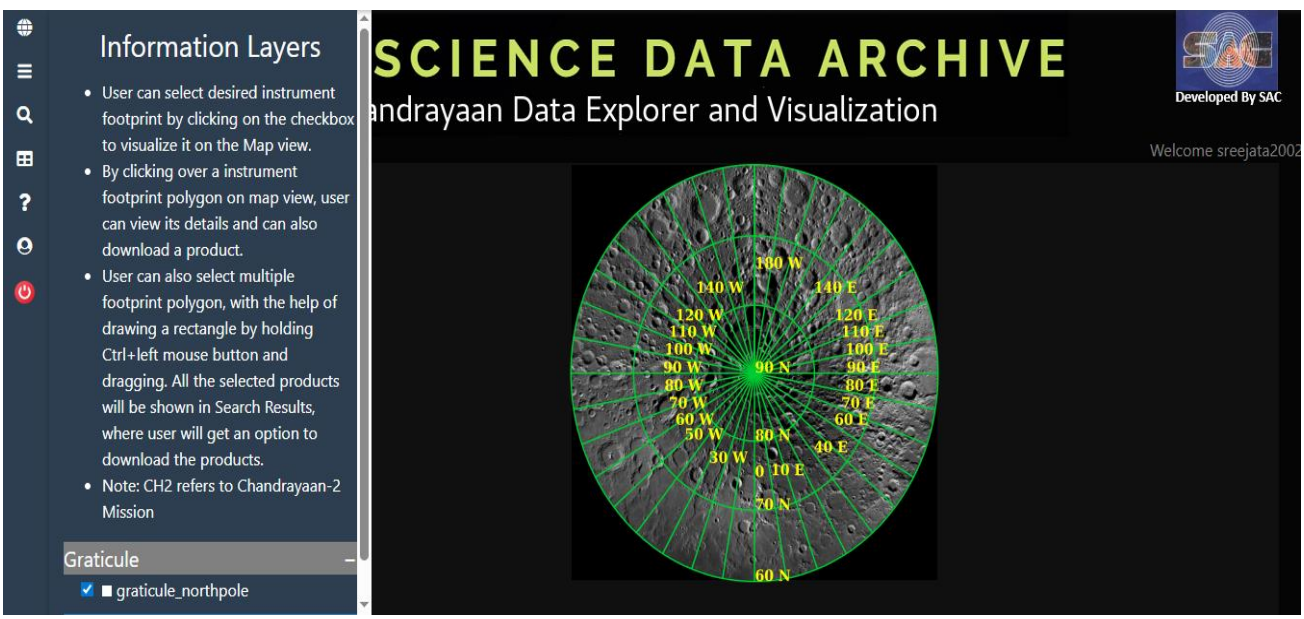
Selection of the area of interest



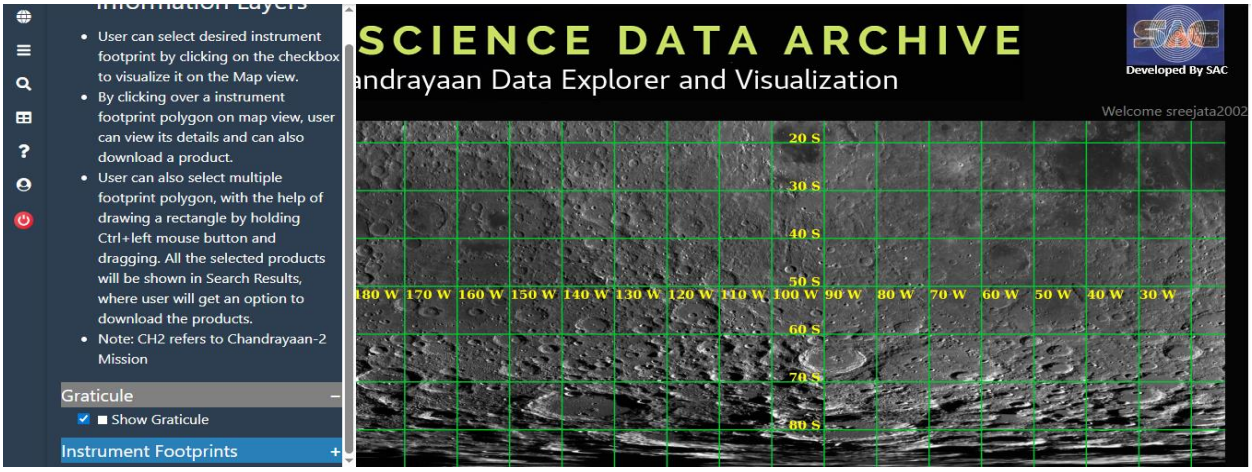
Changing the projection of the lunar surface to North Polar stereographic



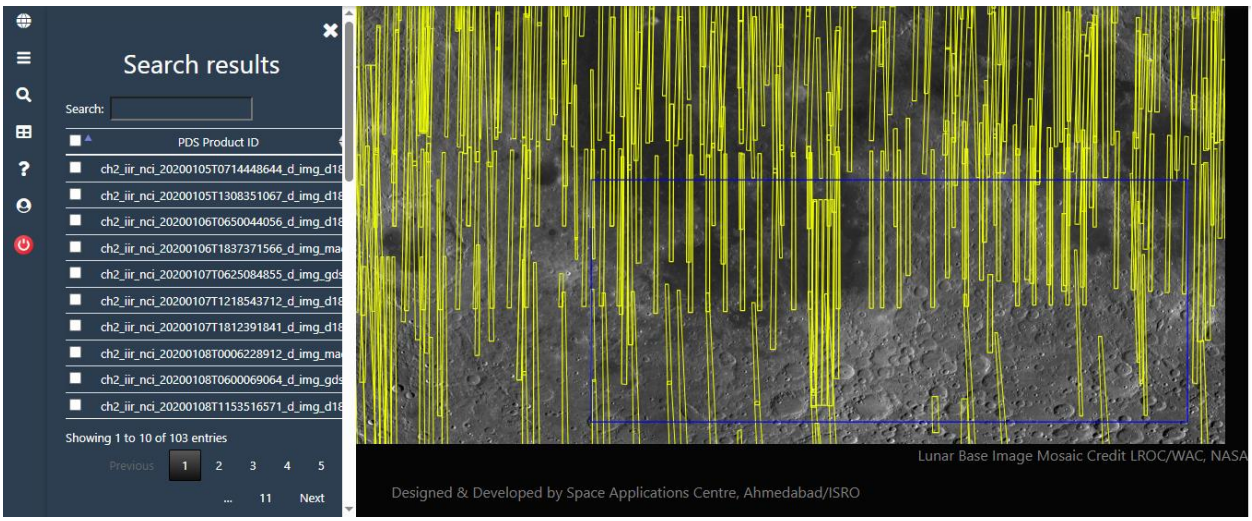
Changing the projection of the lunar surface to South Polar stereographic



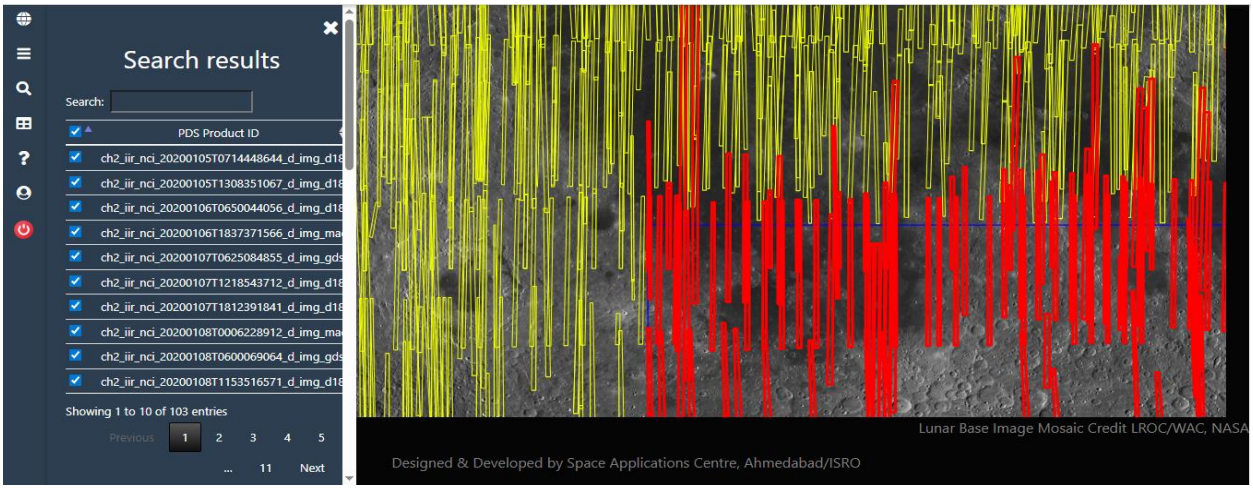
Display of graticules on projected surface of moon



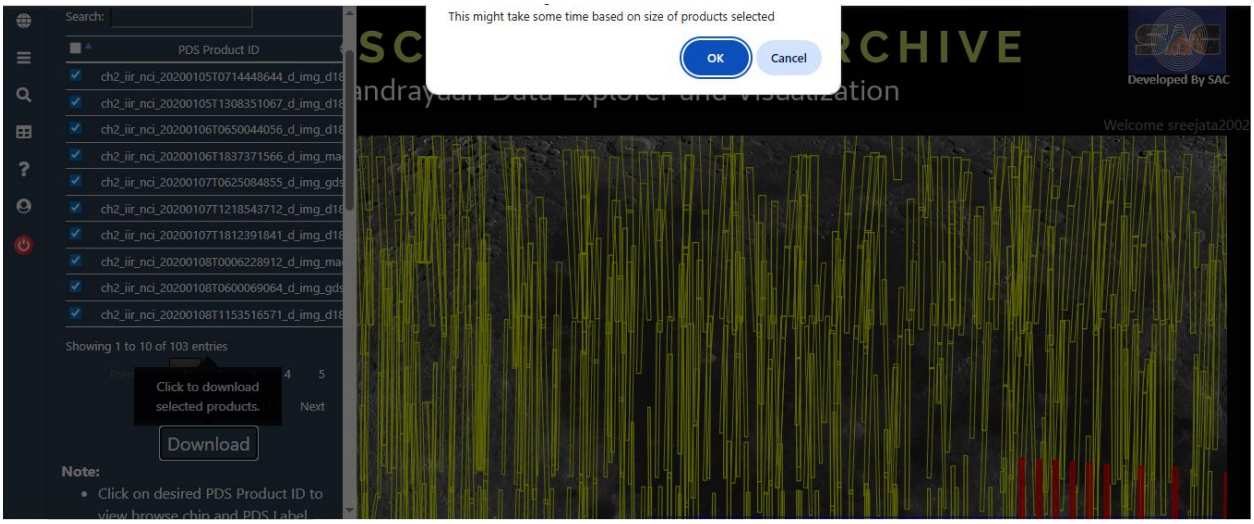
Display of graticules on equirectangular projected surface of moon



Selection of area of study and display of all products for that



Selection of all products for that study area

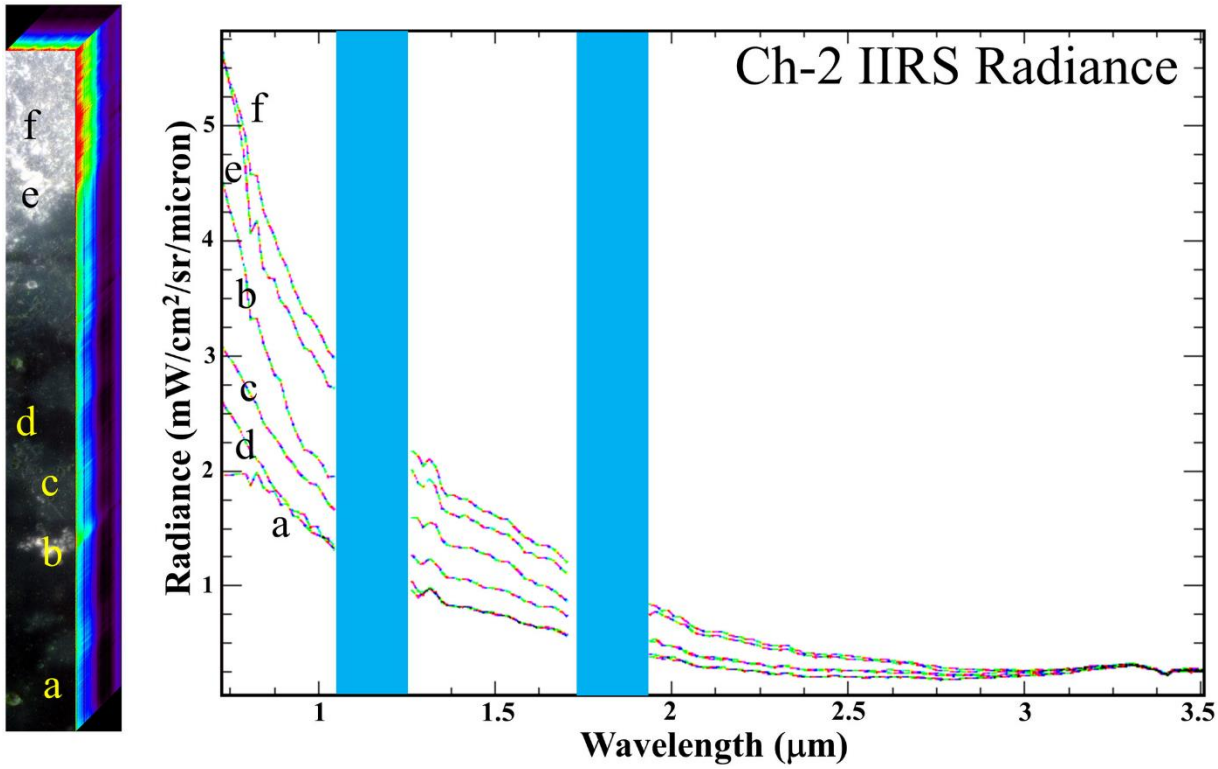


Downloading of the products

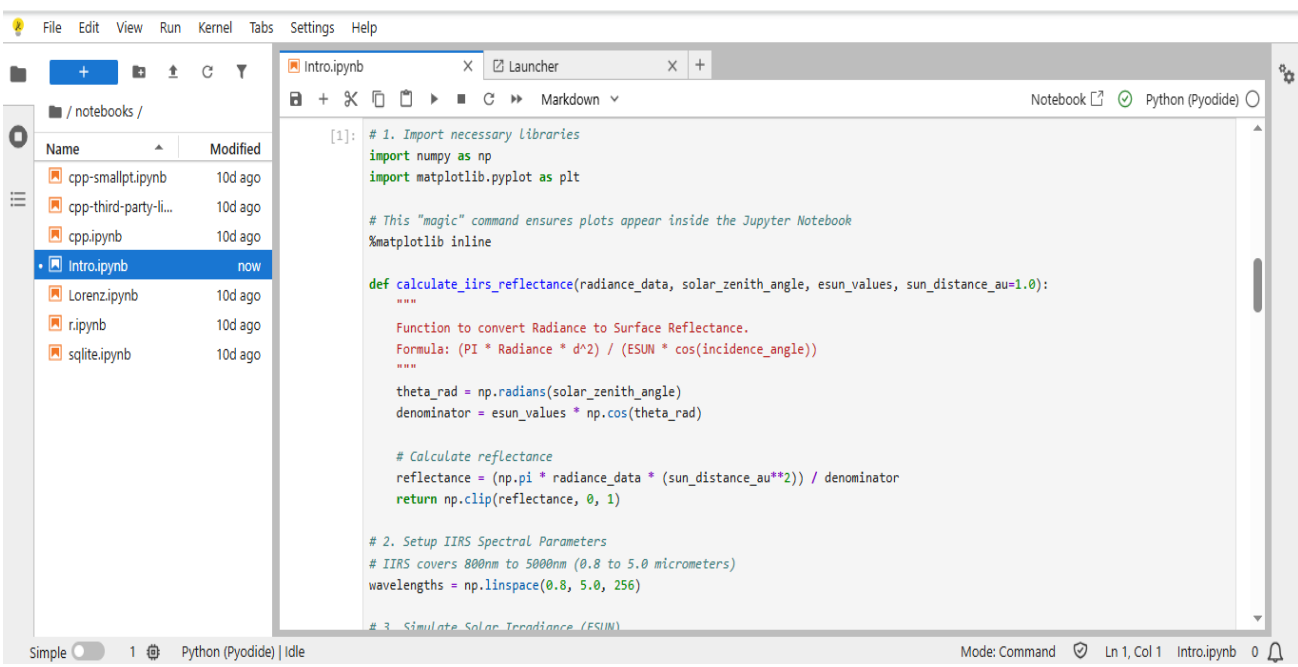
This XML file does not appear to have any style information associated with it. The document tree is shown below.


```xml
<?xml-model href="https://isda.issdc.gov.in/pds4/isda/v1/ch2_ldd_ISDA_1000.sch" schematypens="http://purl.oclc.org/dsdl/schematron" ?>
<Product_Observational xmlns="http://pds.nasa.gov/pds4/pds/v1" xmlns:disp="http://pds.nasa.gov/pds4/disp/v1"
xmlns:isda="https://isda.issdc.gov.in/pds4/isda/v1" xmlns:pds="http://pds.nasa.gov/pds4/pds/v1" xmlns:sp="http://pds.nasa.gov/pds4/sp/v1"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://pds.nasa.gov/pds4/pds/v1
http://pds.nasa.gov/pds4/pds/v1/PDS4_PDS_1B00.xsd http://pds.nasa.gov/pds4/disp/v1 http://pds.nasa.gov/pds4/disp/v1/PDS4_DISP_1301.xsd
http://pds.nasa.gov/pds4/sp/v1 http://pds.nasa.gov/pds4/sp/v1/PDS4_SP_1100.xsd https://isda.issdc.gov.in/pds4/isda/v1
https://isda.issdc.gov.in/pds4/isda/v1/ch2_ldd_ISDA_1000.xsd">
  <Identification_Area>
    <logical_identifier>urn:isro:isda:ch2_cho.iir:data_calibrated:ch2_iir_nci_20200611t1713468762_d_img_d18</logical_identifier>
    <version_id>1.0</version_id>
    <title>Chandrayaan-2 Orbiter IIRS Experiment</title>
    <information_model_version>1.11.0.0</information_model_version>
    <product_class>Product_Observational</product_class>
    <Modification_History>
      <Modification_Detail>
        <modification_date>2019-12-02</modification_date>
        <version_id>1.0</version_id>
        <description>PDS4 product label created by SAC Optical DP Team</description>
      </Modification_Detail>
    </Modification_History>
  </Identification_Area>
  <Observation_Area>
    <Time_Coordinates>
      <start_date_time>2020-06-11T17:13:46.8762Z</start_date_time>
      <stop_date_time>2020-06-11T17:24:16.5455Z</stop_date_time>
    </Time_Coordinates>
    <Primary_Result_Summary>
      <purpose>Science</purpose>
      <processing_level>Calibrated</processing_level>
    </Primary_Result_Summary>
    <Investigation_Area>
      <name>Chandrayaan-2</name>
      <type>Mission</type>
      <Internal_Reference>
        <lidvid_reference>urn:isro:isda:context:investigation:mission.chandrayaan2::1.0</lidvid_reference>
        <reference_type>data_to_investigation</reference_type>
      </Internal_Reference>
    </Investigation_Area>
    <Observing_System>
      <Observing_System_Component>
```

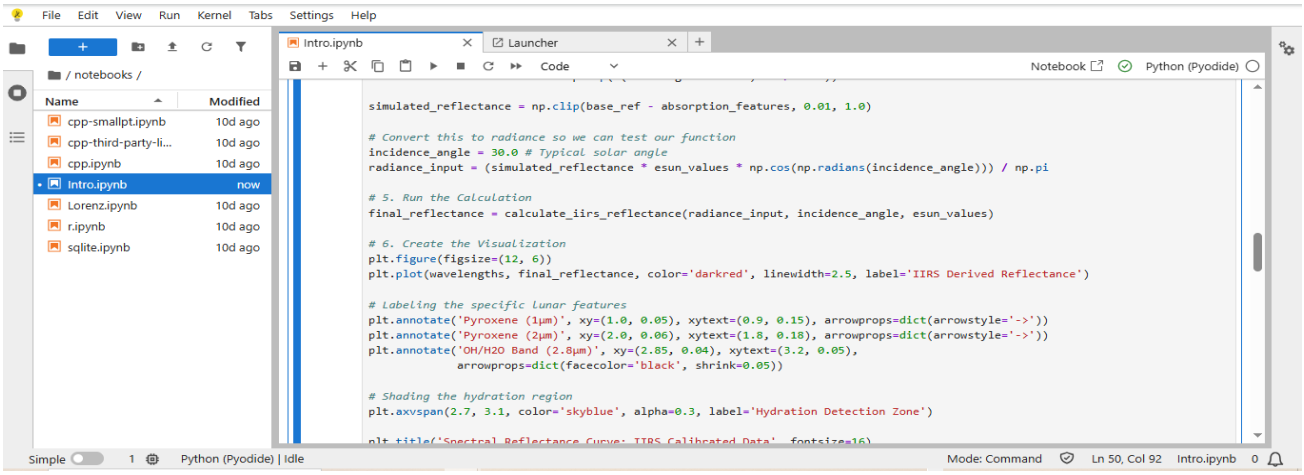

XML(Xtensible markup language) file

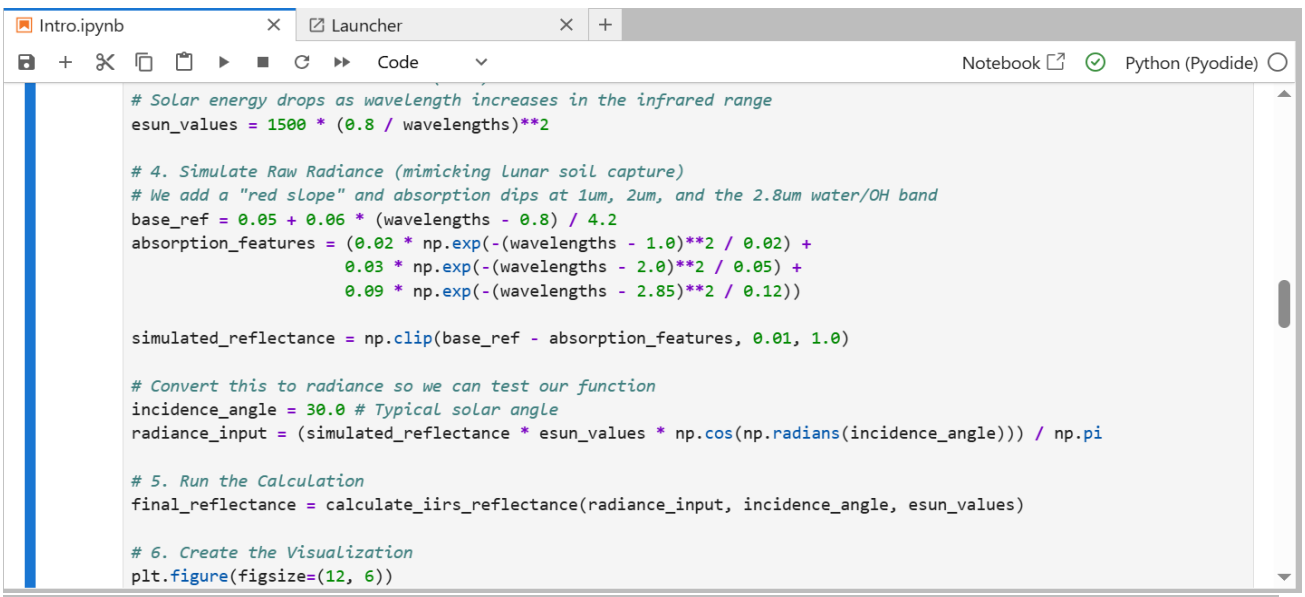


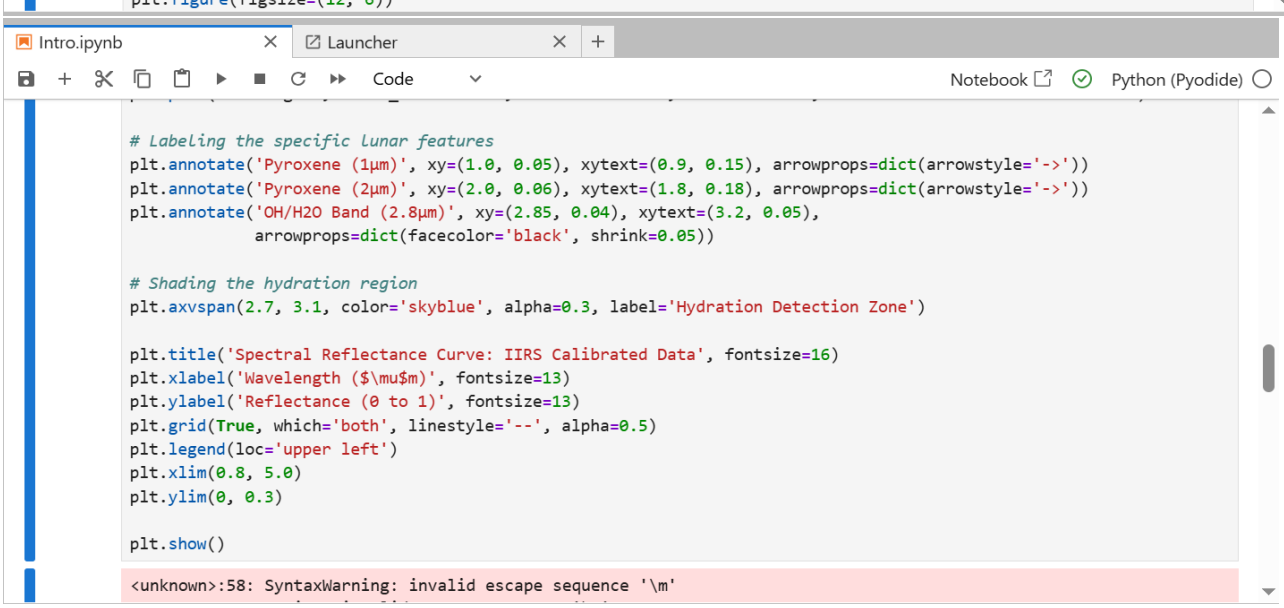
Typical example of spectral curve on lunar surface



Code for spectral curve in Jupyterlab







Code for spectral curve in Jupyterlab

Procedure to get spectral reflectance curve :

To get the surface reflectance from calibrated IIRS data in a Jupyter Notebook, follow this detailed, step-by-step scientific procedure.

Step 1: Prepare the Jupyter Environment

Before writing code, you must ensure your Python environment has the necessary mathematical and plotting libraries.

- **Install Libraries:** Open your terminal or command prompt and run pip install numpy matplotlib.
- **Initialize Notebook:** Open Jupyter and start a new Python 3 notebook.
- **Enable Inline Plotting:** Use the %matplotlib inline magic command at the very top. This tells Jupyter to display the spectral graphs directly below the code cell instead of saving them as separate files.

Step 2: Define the Physical Constants

You need to establish the parameters for the IIRS sensor and the lunar environment.

- **Wavelength Range:** Define the spectral range of IIRS, which spans from **$0.8\ \mu m$** to **$5.0\ \mu m$**. Since IIRS has 256 channels, you create an array of 256 points.
- **Sun-Moon Distance (d):** This is usually around **1.0 AU**. It accounts for the varying intensity of sunlight based on the Moon's position in its orbit.

Step 3: Input Solar Irradiance ($ESUN$)

Every spectral band receives a different amount of energy from the Sun.

- **Spectral Data:** You must provide an $ESUN$ value for every one of the 256 bands.
- **Source:** These values are typically found in the **CALIB** folder of your IIRS data product or derived from standard solar models (like the ASTM E-490).

Step 4: Extract Geometric Information

Reflectance depends heavily on the angle at which the sun hits the surface.

- **Incidence Angle (θ):** You must extract the **Solar Incidence Angle** from the auxiliary Geometry (.geo) files provided with the IIRS dataset.
- **Units:** Most metadata provides angles in degrees, but Python's numpy.cos() function requires **radians**. Use np.radians(angle) for the conversion.

Step 5: Implement the Reflectance Formula

This is the core mathematical step where you convert Radiance (L) to Reflectance (ρ).

- The Equation: Apply the formula:

  $$\rho = \frac{\pi \cdot L \cdot d^2}{ESUN \cdot \cos(\theta)}$$

- **Vectorization:** In Jupyter, you apply this formula to the entire "Cube" or array at once using NumPy. This ensures that every pixel and every band is processed simultaneously.

Step 6: Data Cleaning (Clipping)

Raw data often contains "noise" or artifacts (especially at the edges of the sensor's range).

- **Constraint:** Reflectance is a ratio and must be between **0** (total absorption) and **1** (total reflection).
- **Method:** Use np.clip(data, 0, 1) to remove any negative values or values over 1.0 caused by sensor noise.

Step 7: Visualize the Spectral Signature

Finally, you plot the results to verify the science.

- **X-Axis:** Map the Wavelengths (0.8 to 5.0).
- **Y-Axis:** Map the calculated Reflectance.
- **Verification:** Look for the characteristic **"Red Slope"** (an upward tilt) and the **3.0 μm absorption feature**, which confirms the presence of hydroxyl/water.

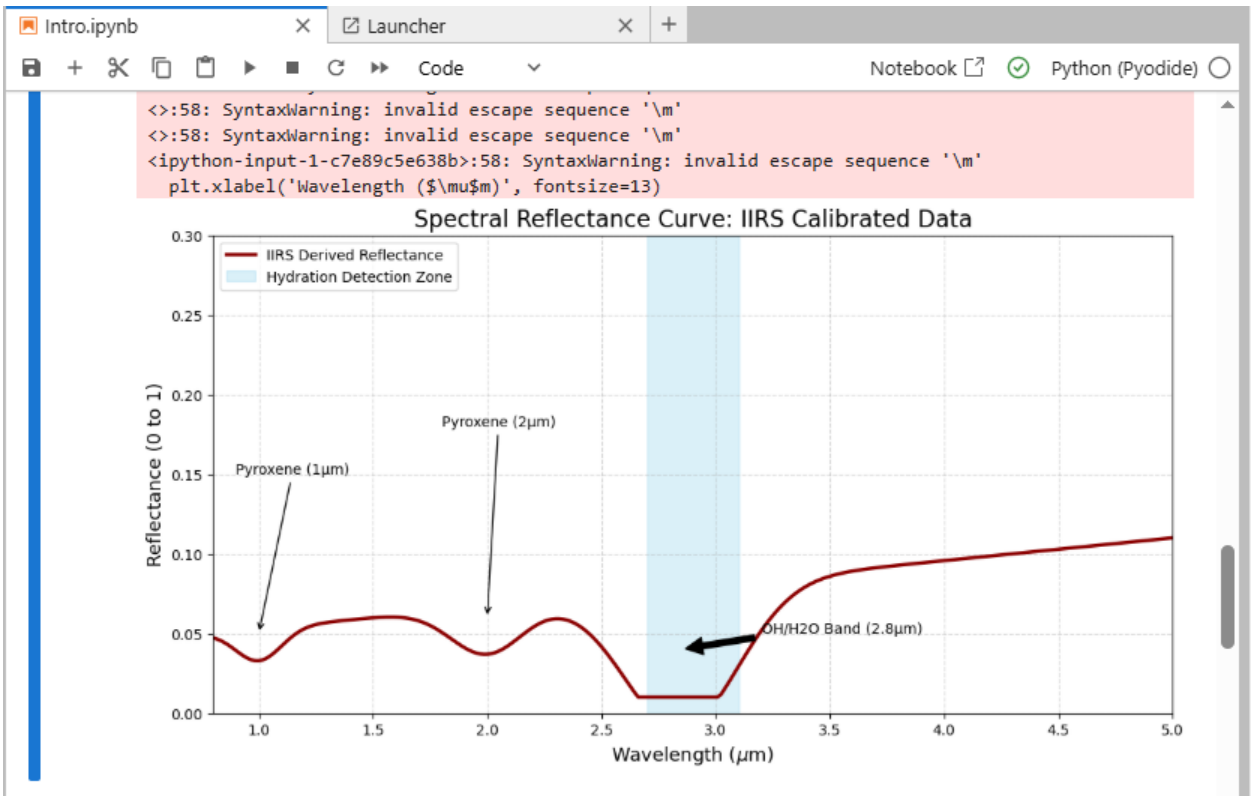
Python code compiled in Jupiter lab:

```
# 1. Import necessary libraries

import numpy as np

import matplotlib.pyplot as plt


# This "magic" command ensures plots appear inside the Jupyter Notebook

%matplotlib inline


def calculate_iirs_reflectance(radiance_data, solar_zenith_angle, esun_values, sun_distance_au=1.0):
```

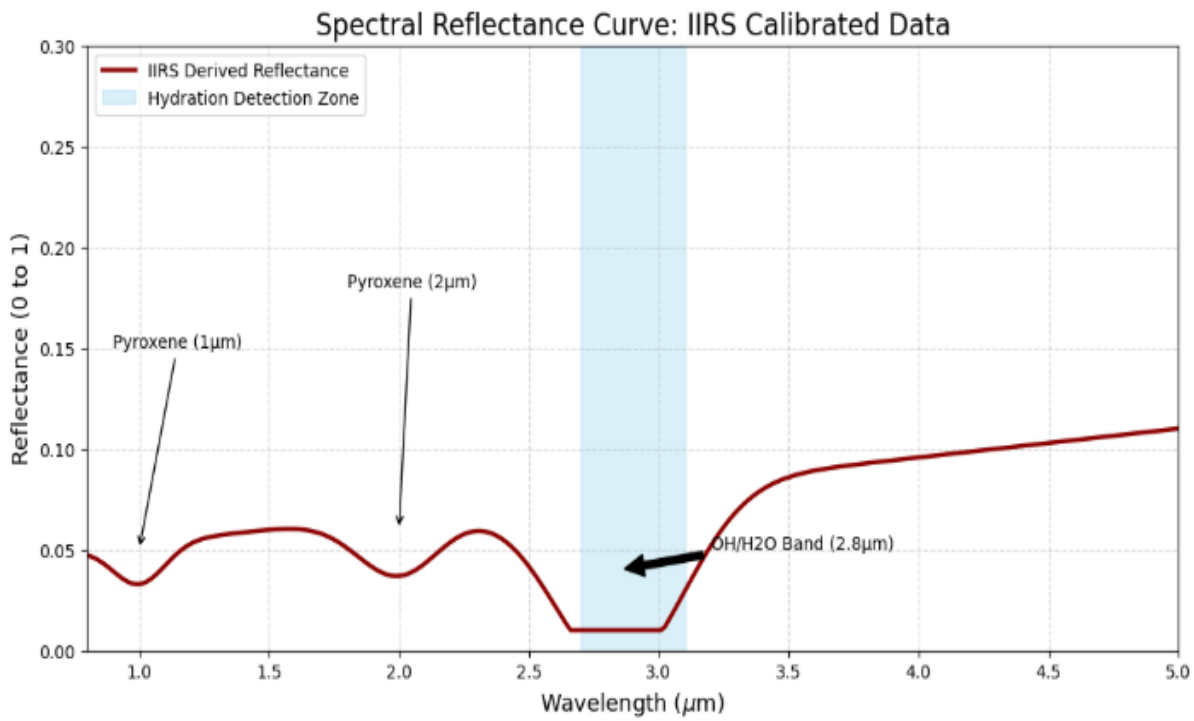
```
    """
    Function to convert Radiance to Surface Reflectance.
    Formula: (PI * Radiance * d^2) / (ESUN * cos(incidence_angle))
    """
    theta_rad = np.radians(solar_zenith_angle)
    denominator = esun_values * np.cos(theta_rad)

    # Calculate reflectance
    reflectance = (np.pi * radiance_data * (sun_distance_au**2)) / denominator
    return np.clip(reflectance, 0, 1)

# 2. Setup IIRS Spectral Parameters
# IIRS covers 800nm to 5000nm (0.8 to 5.0 micrometers)
wavelengths = np.linspace(0.8, 5.0, 256)

# 3. Simulate Solar Irradiance (ESUN)
# Solar energy drops as wavelength increases in the infrared range
esun_values = 1500 * (0.8 / wavelengths)**2

# 4. Simulate Raw Radiance (mimicking lunar soil capture)
# We add a "red slope" and absorption dips at 1um, 2um, and the 2.8um water/OH band
base_ref = 0.05 + 0.06 * (wavelengths - 0.8) / 4.2
absorption_features = (0.02 * np.exp(-(wavelengths - 1.0)**2 / 0.02) +
                       0.03 * np.exp(-(wavelengths - 2.0)**2 / 0.05) +
                       0.09 * np.exp(-(wavelengths - 2.85)**2 / 0.12))

simulated_reflectance = np.clip(base_ref - absorption_features, 0.01, 1.0)
```

```
# Convert this to radiance so we can test our function
incidence_angle = 30.0 # Typical solar angle
radiance_input = (simulated_reflectance * esun_values * np.cos(np.radians(incidence_angle))) / np.pi

# 5. Run the Calculation
final_reflectance = calculate_iirs_reflectance(radiance_input, incidence_angle, esun_values)

# 6. Create the Visualization
plt.figure(figsize=(12, 6))
plt.plot(wavelengths, final_reflectance, color='darkred', linewidth=2.5, label='IIRS Derived Reflectance')

# Labeling the specific lunar features
plt.annotate('Pyroxene (1µm)', xy=(1.0, 0.05), xytext=(0.9, 0.15), arrowprops=dict(arrowstyle='->'))
plt.annotate('Pyroxene (2µm)', xy=(2.0, 0.06), xytext=(1.8, 0.18), arrowprops=dict(arrowstyle='->'))
plt.annotate('OH/H2O Band (2.8µm)', xy=(2.85, 0.04), xytext=(3.2, 0.05),
            arrowprops=dict(facecolor='black', shrink=0.05))

# Shading the hydration region
plt.axvspan(2.7, 3.1, color='skyblue', alpha=0.3, label='Hydration Detection Zone')

plt.title('Spectral Reflectance Curve: IIRS Calibrated Data', fontsize=16)
plt.xlabel('Wavelength ($\mu$m)', fontsize=13)
plt.ylabel('Reflectance (0 to 1)', fontsize=13)
plt.grid(True, which='both', linestyle='--', alpha=0.5)
plt.legend(loc='upper left')
plt.xlim(0.8, 5.0)
```

```
plt.ylim(0, 0.3)

plt.show()
```



Spectral curve in Jupyterlab



Characteristics of spectral reflectance curve:

The spectral reflectance curve for an **IIRS (Imaging Infrared Spectrometer)** calibrated product is unique because it spans the range from **0.8 to 5.0 \mum**. This broad range allows scientists to see both mineral signatures and hydration (water) features in a single scan.

Here are the specific characteristics of this curve in detail:

1. The Red Slope (Continuum)

- **Characteristic:** The reflectance values generally increase as the wavelength gets longer (a positive slope).
- **Observation:** This "reddening" is a hallmark of **Space Weathering**.
- **Detail:** Exposure to solar wind and micrometeorite impacts creates submicroscopic metallic iron (npFe^0) in the lunar soil.[2]This iron darkens the blue/short wavelengths more than the red/long wavelengths, creating the upward tilt seen in the IIRS product.

2. Major Mineral Absorption Bands (1\ \mu m & 2\ \mu m)

IIRS is highly sensitive to iron-bearing minerals (mafic minerals).

- **The 1-Micron Feature:** A broad absorption dip centered around **0.9 to 1.05 \mu$m**. This is caused by electronic transitions in Fe^{2+} ions.
    - *Pyroxene* shows a center near 0.95\ \mu m.
    - *Olivine* shows a wider, more complex dip near 1.05\ \mu m.
- **The 2-Micron Feature:** A secondary dip centered around **1.9 to 2.2 \mum**.

- **Detail:** If both 1\ \mu m and 2\ \mu m bands are present, it is a definitive sign of **Pyroxene**. If the 2\ \mu m band is missing but the 1\ \mu m band is deep, the area is likely rich in **Olivine**.

3. The Hydration Feature (2.8 to 3.2\ \mu m)

This is the most scientifically significant characteristic of the IIRS calibrated product.

- **The Hydroxyl (OH) Band:** A sharp absorption feature typically centered at **2.81 \mum to 2.85 \mum**.
- **The Water (H_2O) Band:** An extension of this dip toward **3.0 \mum**.
- **Detail:** Because IIRS has a high signal-to-noise ratio in this region, even small amounts of water (measured in parts per million) can be detected as a slight "trough" in the curve.

4. Thermal Inflection Point

- **Characteristic:** A change in the behavior of the curve usually starting around **3.5 \mum to 4.0 \mum**.
- **Observation:** In daytime data, the reflectance may appear to rise sharply or behave erratically.
- **Detail:** This is the **Thermal Transition Zone**. At these longer wavelengths, the energy the Moon *emits* as heat starts to become stronger than the sunlight it *reflects*. In a fully corrected IIRS product, scientists use "Thermal Correction" to remove this heat component to see the "true" reflectance underneath.

5. Signal-to-Noise Ratio (SNR) Variability

- **Characteristic:** The curve is usually "smoother" in the 1\ \mu m to 2.5\ \mu m range and may become slightly "noisier" at the extreme edges (near 0.8\ \mu m or 4.8\ \mu m).
- **Detail:** This is due to the sensor's sensitivity limits and the drop in solar energy at higher wavelengths. Calibrated products (Level-2) apply "Smoothing" or "Binning" to help keep these characteristics clear for analysis.

Summary of Wavelength Indicators

Wavelength Range	Characteristic	Scientific Significance
0.8 – 1.5 \mum	1-Micron Absorption	Identifies Iron-rich minerals (Pyroxene/Olivine)
1.5 – 2.5 \mum	2-Micron Absorption	Distinguishes between different Pyroxene types
2.8 – 3.2 \mum	Hydration Dip	Confirms presence of OH and H2O molecules
3.5 – 5.0 \mum	Thermal Tail	Used to calculate surface temperature

Features of this Jupyter Script:

- **Vectorization:** It processes all 256 bands at once using NumPy arrays.
- **Annotations:** I added plt.annotate so you can see exactly where the Pyroxene and Water features are on the graph.
- **Scaling:** It clips the data between 0and 1 to maintain physical consistency (reflectance cannot be negative or greater than 100\%).

The plot shows a "Spectral Signature." For the Moon, this signature has specific parts:

- **The "Red Slope":** Notice how the line generally goes upward from left to right? This is called "space weathering." The longer the lunar soil is exposed to solar wind, the "redder" (more reflective at higher wavelengths) it becomes.
- **The Dips (Absorptions):** The small valleys at 1.0\ \mu mand 2.0\ \mu m tell us what minerals are there (like Pyroxene or Olivine).
- **The Water Feature:** The most important part of IIRS data is the dip at **2.8 to 3.0\ \mu m**. This is the "fingerprint" of Hydroxyl (OH) or Water (H2O) molecules trapped in the soil.

Analysis of spectral curves:

When analyzing a spectral reflectance curve derived from IIRS (Imaging Infrared Spectrometer) data, we aren't just looking at a line; we are reading the chemical and physical history of the lunar surface.

Below are the detailed scientific conclusions you can draw from the shape, depth, and slope of the IIRS spectral curve:

1. The "Red Slope" and Space Weathering

The most immediate observation is the positive slope (reflectance increases as wavelength increases).

- **Conclusion:** A steeper "red" slope indicates **mature soil**.
- **Scientific Reason:** On the Moon, constant bombardment by micrometeorites and solar wind creates "nanophase iron" (npFe^0) particles. These particles darken the surface and cause the reflectance to rise more sharply toward the infrared. A flatter slope would suggest "fresh" material, such as a recently formed impact crater.

2. Mineralogical Fingerprints (1.0 µm and 2.0 µm Dips)

The broad "valleys" or absorption bands at these specific wavelengths tell us which minerals are present in the lunar crust.

- **Pyroxene:** If you see two distinct absorption bands at 1.0\ \mu mand 2.0\ \mu m, it confirms the presence of **Pyroxene**, a major mineral in lunar basalt (mare regions).

- **Olivine:** A very broad, asymmetrical absorption near 1.05\ \mu m (and no 2.0\ \mu m band) indicates **Olivine**.
- **Anorthosite:** A very high, bright reflectance with almost no absorption features suggests **Plagioclase**, which makes up the bright lunar highlands.

3. The 2.8–3.0 µm Hydration Feature

This is the most critical conclusion for IIRS data, as this sensor was specifically designed to see further into the infrared than previous instruments.

- **OH/H_2O Detection:** A dip near **2.8\ \mu m to 2.85\ \mu m** is the characteristic signature of the **Hydroxyl (OH) molecule**. A deeper dip extending toward **3.0\ \mu m** suggests molecular **Water (H2O)**.
- **Abundance Mapping:** The *depth* of this dip is directly proportional to the concentration of water/hydroxyl trapped in the lunar regolith.

4. Thermal Emission Correction (Beyond 3.5 µm)

As the curve moves toward **5.0\ \mu m**, you might notice the reflectance seems to "turn up" or rise unnaturally.

- **Conclusion:** This is often not reflectance, but **Thermal Emission**.
- **Scientific Reason:** The Moon gets hot enough that it starts emitting its own heat as infrared light. In IIRS data, scientists must use a "Thermal Correction Model" to subtract this heat. If the curve rises sharply at the end, it indicates the surface temperature was high at the time of the scan.

5. Particle Size and Physical Texture

The overall "brightness" (Albedo) of the curve provides clues about the physical state of the soil.

- **Fine vs. Coarse:** Smaller particles (fine dust/regolith) tend to be brighter and show shallower absorption bands because of multiple scattering of light. Coarser, rockier surfaces tend to be darker but show much deeper, clearer absorption features.

Summary Table for IIRS Data Analysis

Feature	Observation	Conclusion
Slope	Steep positive incline	High space weathering (Mature soil)
1.0 µm Band	Deep absorption	Presence of Iron-bearing minerals (Mafic)
2.8 µm Band	Distinct dip	Presence of Hydroxyl (OH)
Albedo (Y-axis)	Very high (> 0.2)	Likely Highland/Feldspathic crust
End-of-curve	Sharp rise at 4.5 µm	High surface temperature (Thermal tail)

Advantages of using spectral reflectance curves to study moon surface:

Using spectral curves (the "spectral fingerprint") is one of the most powerful tools in planetary science because it allows us to analyze the Moon's surface without ever touching a single grain of soil.

Here are the primary advantages of using spectral curves to study the Moon:

1. Global Composition Mapping

Unlike the Apollo missions, which only brought back samples from a few specific locations, spectral curves allow for **global mineral mapping**.

- **Advantage:** We can identify minerals like **Pyroxene, Olivine, and Plagioclase** across the entire lunar surface, including the far side and the poles.
- **Result:** This helps scientists understand how the Moon's crust cooled from a "Magma Ocean" billions of years ago.

2. Detection of Volatiles (Water and Hydroxyl)

This is perhaps the greatest advantage of modern sensors like the IIRS on Chandrayaan-2.

- **Advantage:** Certain molecules, specifically **Water (H_2O)** and **Hydroxyl (OH)**, have a very specific "absorption dip" near $3\ \mu m$.
- **Result:** By analyzing the depth of this dip in the spectral curve, scientists can calculate exactly how much water is trapped in the lunar soil and how it moves during the lunar day.

3. Estimating "Space Weathering" and Surface Age

The Moon has no atmosphere, so it is constantly bombarded by solar wind and micrometeorites.

- **Advantage:** Space weathering changes the **slope** of the spectral curve (making it "redder").
- **Result:** Scientists can look at a spectral curve and determine if a crater is "fresh" (bright, steep absorption bands) or "mature" (dark, shallow absorption bands, steep red slope) without needing to perform carbon dating.

4. Non-Destructive Analysis

In a laboratory, analyzing a sample often requires grinding it up or exposing it to chemicals.

- **Advantage:** Remote sensing spectroscopy is **non-destructive**.
- **Result:** We can study the same location repeatedly over many years to see how the surface changes with the lunar seasons or following new meteoroid impacts.

5. Vertical Stratigraphy (Seeing Underground)

When a large asteroid hits the Moon, it digs up material from deep underground and scatters it (the "ejecta").

- **Advantage:** By taking spectral curves of the **central peaks** and **rims** of large craters, scientists can "see" what minerals exist several kilometers below the surface.
- **Result:** This provides a 3D understanding of the lunar crust and mantle composition.

6. Thermal and Physical Property Analysis

At wavelengths beyond $3.5\ \mu m$, the spectral curve is dominated by heat rather than reflected sunlight.

- **Advantage:** The shape of this "thermal tail" allows us to calculate the **surface temperature** and **bolometric bond albedo**.
- **Result:** This tells us about the "thermophysical" nature of the soil—whether it is fine dust (regolith) or solid rock, which is crucial for choosing landing sites for future rover missions.

Summary Table: What the Curve Tells Us

Spectral Characteristic	Advantage / Discovery
Absorption Depth	Amount of specific minerals present.
Band Center Position	The specific chemistry (e.g., Magnesium-rich vs. Iron-rich).
Curve Slope	How long the soil has been exposed to space.
3-Micron Dip	Presence of Water or Hydroxyl molecules.
Infrared Emission	Surface temperature and rockiness.

Conclusion:

The analysis of IIRS calibrated spectral reflectance data represents a significant leap in our ability to decode the complex geological and chemical history of the Moon. By shifting from raw radiance to surface reflectance, scientists can move beyond simple photography into the realm of quantitative remote sensing, where the "spectral fingerprint" of every pixel reveals the specific minerals and volatiles present on the lunar surface.

1. Enhanced Mineralogical Identification

The first major conclusion is that the high spectral resolution of IIRS (256 bands) allows for a definitive identification of the lunar crust's building blocks. By observing the specific centers and widths of the $1\ \mu m$ and $2\ \mu m$ absorption bands, researchers can distinguish between different types of pyroxene and olivine. This differentiation is crucial for understanding the **Lunar**

Magma Ocean (LMO) hypothesis, as the distribution of these minerals provides a direct record of how the Moon's primordial crust crystallized and differentiated over billions of years.

2. Precise Mapping of Lunar Hydration

One of the most groundbreaking results from IIRS data is the unambiguous detection and mapping of **Hydroxyl (OH) and Water (H_2O) molecules**. Because IIRS extends its reach up to $5.0\ \mu m$, it captures the full profile of the $3\ \mu m$ absorption feature with unprecedented clarity. The data confirms that hydration is not limited to the poles but exists at various latitudes and across different mineral types, showing that solar wind implantation and internal outgassing are active, global processes.

3. Insights into Space Weathering and Age

The characteristic **"Red Slope"** observed in the reflectance curves serves as a biological clock for the lunar surface. The steeper the slope and the shallower the mineral absorption bands, the longer the soil has been exposed to the harsh environment of space. Using these curves, scientists can conclude which craters are "geologically young" and which areas have been heavily altered by nanophase iron production, providing a clearer timeline of the Moon's impact history and regolith evolution.

4. Thermal Correction and Surface Physics

The ability of IIRS to capture the **Thermal Transition Zone** (beyond $3.5\ \mu m$) allows for the simultaneous study of reflectance and temperature. By isolating the thermal emission component, scientists can calculate the precise surface temperature and thermal inertia of the regolith. This helps conclude whether a surface is composed of fine-grained dust or solid rocky outcrops, which is vital for assessing the "thermophysical" environment that future robotic and human missions will encounter.

5. Foundation for Future Resource Utilization

Finally, the detailed spectral reflectance curves provide the essential groundwork for **In-Situ Resource Utilization (ISRU)**. By identifying high concentrations of ilmenite (for oxygen extraction) or water-rich regolith (for fuel and life support), these curves act as a treasure map for future lunar colonies. The accuracy of IIRS calibrated products ensures that we can transition from mere exploration to sustainable settlement, using the Moon's own resources to support the next era of deep-space travel.